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# Spring Boot 4.0 Release Notes

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## Upgrading from Spring Boot 3.5

Since this is a major release of Spring Boot, upgrading existing applications can be a little more involved that usual. We've put together a dedicated migration guide to help you upgrade your existing Spring Boot 3.5 applications.

If you're currently running with an earlier version of Spring Boot, we strongly recommend that you upgrade to Spring Boot 3.5 before migrating to Spring Boot 4.0.

## New and Noteworthy

| Tip | Check the configuration changelog for a complete overview of the changes in configuration. |
|---|---|

### Milestones Available from Maven Central

Starting with 4.0.0-M1, all Spring Boot milestones (and release candidates) are now published to Maven Central in addition to https://repo.spring.io. This should make it easier to try new milestones in the 4.x line as they become available.

### Gradle 9

Gradle 9 is now supported for building Spring Boot applications. Support for Gradle 8.x (8.14 or later) remains.

### HTTP Service Clients

Spring Boot now includes auto-configuration support and configuration properties for HTTP Service Clients. HTTP Service Clients allow you to annotate plain Java interfaces and have Spring automatically create implementations of them.

For example, the following interface can be used to call an "echo" service:

```
@HttpExchange(url = "https://echo.zuplo.io")
public interface EchoService {

    @PostExchange
    Map<?, ?> echo(@RequestBody Map<String, String> message);

}
```

For full details of the feature, please see the updated documentation.

### API Versioning

Auto-configuration has been added for API Versioning with Spring MVC and with Spring WebFlux.

API Versioning can be configured using the `spring.mvc.apiversion.*` or `spring.webflux.apiversion.*` properties. For more advanced control, beans of type `ApiVersionResolver`, `ApiVersionParser` and `ApiVersionDeprecationHandler` can be defined.

See the reference documentation (MVC, WebFlux) for further details.

### JmsClient

The auto-configuration for JMS now includes support for the new `JmsClient` API. The support for `JmsTemplate` and `JmsMessagingTemplate` is left unchanged.

### Task Decoration

The auto-configurations for task scheduling and task execution now support multiple `TaskDecorator` beans. When the context contains multiple `TaskDecorator` beans, a `CompositeTaskDecorator` that delegates to them is created. The individual decorators are called in the ordered defined by `@Order` and `Ordered`.

### OpenTelemetry starter

A new starter, `spring-boot-starter-opentelemetry` has been added. This starter brings in all necessary dependencies to export metrics and traces over OTLP. It will also auto-configure the OpenTelemetry SDK.

### Configuration Properties Metadata for External Types

It is now possible for `@ConfigurationProperties`-annotated types to refer to types that are located in a different module. To source the metadata from those modules, you should add the annotation processor (if necessary) and flag the type with `@ConfigurationPropertiesSource`.

See the reference documentation for more details.

### SSL Info

Support for the certificate validity threshold has been removed from the SSL info contribution. A certificate that had a status of `WILL_EXPIRE_SOON` will now appear as `VALID`. The information about the start and end of a certificate's validity remains.

### SSL Health

Certificate chains that contain one or more certificates that will expire within the configured threshold (`management.health.ssl.certificate-validity-warning-threshold`) are now listed in a new `expiringChains` entry in the details of the health response. The status `WILL_EXPIRE_SOON` is no longer used and expiring certificates will have a status of `VALID`.

### MongoDB Health Indicators

The MongoDB health indicators have been reworked so that they no longer require Spring Data MongoDB. This allows health information to be provided when using the MongoDB Java Driver directly.

As part of this change, the health indicators have moved from `spring-boot-data-mongodb` to `spring-boot-mongodb`. Their packages have also been updated accordingly.

#### MongoDB Properties

A new property, `spring.data.mongodb.representation.big-decimal`, has been introduced to control how Spring Data MongoDB stores `BigDecimal` (and `BigInteger`) values in MongoDB.

A number of properties have also been renamed. See the migration guide for details.

### Kotlin Serialization

Spring Boot now ships a new "spring-boot-kotlinx-serialization-json" module and corresponding "spring-boot-starter-kotlin-serialization" for Kotlin Serialization support. This will contribute a `Json` bean and configure it with the available `spring.kotlinx.serialization.json.*` properties. Note that a `HttpMessageConverter` will be also contributed to the application and will be set ahead of other JSON converters (acting as fallbacks).

### RestTestClient

Support for the newly introduced RestTestClient has been added.

With a regular `@SpringBootTest` or when `@AutoConfigureMockMvc` is used, you can autowire a `RestTestClient` that operates on the underlying `MockMvc` instance.

For integration tests, i.e. `@SpringBootTest` with either a defined or random port, a `RestTestClient` can be injected to target the running server.

### Redis Static Master/Replica

Auto-configuration for Static Master/Replica has been added. This feature is only supported by Lettuce.

To use it, provide the list of static nodes using the new `spring.data.redis.masterreplica.nodes` property.

### Redis Observability

The Redis auto-configuration has been improved to auto-configure `MicrometerTracing`, rather than `MicrometerCommandLatencyRecorder`. The former operates on the Observation API and provides both metrics and spans.

### Dependency Upgrades

Spring Boot 4.0 moves to new versions of several Spring projects:

- Micrometer 1.16
- Micrometer Tracing 1.6
- Reactor 2025.0
- Spring AMQP 4.0
- Spring Batch 6.0
- Spring Data 2025.1
- Spring Framework 7.0
- Spring GraphQL 2.0
- Spring HATEOAS 3.0
- Spring Integration 7.0
- Spring for Apache Kafka 4.0
- Spring LDAP 4.0
- Spring for Apache Pulsar 2.0
- Spring REST Docs 4.0
- Spring Security 7.0
- Spring Session 4.0
- Spring WS 5.0

Numerous third-party dependencies have also been updated, some of the more noteworthy of which are the following:

- Artemis 2.43.0
- Brave 6.3
- Commons Codec 1.19
- Commons Lang3 3.19.0
- Couchbase Client 3.9
- Ehcache3 3.11.1
- Elasticsearch Client 9.1
- Flyway 11.11
- GraphQL 24
- Groovy 5.0
- Gson 2.13.2
- H2 2.4
- Hibernate 7.1
- Hibernate Validator 9.0
- HikariCP 7.0
- HtmlUnit 4.16.0
- Jackson 3.0
- Jakarta Annotation 3.0
- Jakarta Persistence 3.2
- Jakarta Servlet 6.1
- Jakarta Validation 3.1
- Jakarta WebSocket 2.2
- Jakarta WS RS 4.0
- Jedis 6.2.0
- Json-smart 2.6.0
- Jetty 12.1
- Kafka 4.1.0
- Kotlin 2.2.20
- Kotlin Serialization 1.9
- Lettuce 6.8.1.RELEASE
- Liquibase 5.0
- Maven AntRun Plugin 3.2
- Maven Javadoc Plugin 3.12
- Mockito 5.20
- MongoDB 5.6.0
- MSSQL JDBC 13.2
- Native Build Tools Plugin 0.11
- Neo4j Java Driver 6.0
- OpenTelemetry 1.54.0
- Pulsar 4.1.0
- R2DBC H2 1.1
- R2DBC Postgresql 1.1
- Rabbit AMQP Client 5.27
- Selenium 4.37
- Selenium HtmlUnit 4.35
- SnakeYAML 2.5
- TestContainers 2.0
- Tomcat 11.0
- XML Maven Plugin 1.2

### Miscellaneous

Apart from the changes listed above, there have also been lots of minor tweaks and improvements including:

- The auto-configuration for Micrometer's metrics aspects now enables support for `@MeterTag` on `@Counted` and `@Timed` methods with a SpEL-based `ValueExpressionResolver`.
- `@ServiceConnection` support for MongoDB now supports Testcontainers' `MongoDBAtlasLocalContainer`.
- Improved error message when configuration property binding fails due to a class not found error.
- Names of several reactive auto-configuration classes have been updated for consistency.
- Public members (aside from constants) have been removed from auto-configuration classes. Auto-configurations never have been public API and now this is enforced through Java mechanisms.
- Auto-configured HTTP clients backed by the JDK `HttpClient` are now configured to use virtual threading when `spring.threads.virtual.enabled` is `true`.
- A `logging.console.enabled` property has been introduced. Setting it to `false` disables console logging.
- Log4j Core configuration file detection for Log4j 3 has been improved.
- Resource lookup in DevTools restart has been optimized.
- `ScheduledTasksObservabilityAutoConfiguration` has been renamed to `ScheduledTasksObservationAutoConfiguration` for consistency.
- Authenticating with Elasticsearch using an API key is now supported using the new `spring.elasticsearch.api-key` property.
- `ConditionalOnEnabledTracing` has been renamed to `ConditionalOnEnabledTracingExport`.
- The property `management.tracing.enabled` has been renamed to `management.tracing.export.enabled`.
- `SanitizableData.getKey()` and `SanitizableData.getLowerCaseKey()` never return `null`. Passing a `null` value as a `key` in the `SanitizableData` will now throw an exception.
- Specialized interfaces have been introduced in `PropertiesConfigAdapter` for better nullability handling.
- The property `spring.dao.exceptiontranslation.enabled` has been renamed to `spring.persistence.exceptiontranslation.enabled`.
- The max size of Tomcat's static cache can now be configured using the `server.tomcat.resource.cache-max-size` property.
- You can now set `Spring-Boot-Jar-Type` to `development-tool` in your `MANIFEST.MF` to exclude dependencies from uber jars.
- Amazon Web Services (AWS) Elastic Container Service (ECS) platform is now a recognized `CloudPlatform`.
- Support for Micrometer's `@ObservationKeyValue` has been added.
- Public members from configurations imported by auto-configuration classes have been made package-private.
- HTML Unit `LocalHostWebClient` and `LocalHostWebConnectionHtmlUnitDriver` have been superseded by `UriBuilderFactoryWebClient` and `UriBuilderFactoryWebConnectionHtmlUnitDriver`.
- A new configuration property can be used to control whether the Elasticsearch Client's sniffer is enabled.

## Deprecations in Spring Boot 4.0.0

- `OperationMethod(Method method, OperationType operationType)` in favor of `OperationMethod(Method method, OperationType operationType, Predicate<Parameter> optionalParameters)`.
- Jackson 2 support ships in a deprecated form.
- `org.springframework.boot.env.EnvironmentPostProcessor` has replaced by `org.springframework.boot.EnvironmentPostProcessor` but the old interface remains in a deprecated form to ease upgrade pain.

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